Rebecca Ann Jackson

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EDUCATION

Villanova University - Villanova, PA
- Master of Science in Computer Science — **GPA: 4.00** | Aug 2025 - May 2027
- Bachelor of Science in Computer Science — **GPA: 3.99** | Aug 2023 - Dec 2026
- Bachelor of Science in Mathematics — **GPA: 3.99** | Aug 2023 - May 2026

Relevant Courses: Machine Learning, Data Science, Computer Systems, Programming Languages, Algorithms & Data Structures

WORK EXPERIENCE

Undergraduate Research Assistant - Villanova, PA — Jan 2024 - Present
- Contributed to 4 projects through Python data analysis, API integration, and web development resulting in 2 co-authored peer-reviewed publications (*Mathematics Magazine*; *NordiCHI 2024 Proceedings*)

Teaching Assistant (TA) - Villanova, PA — Jan 2025 - Present
- Served as TA for Competitive Programming and Data Structures and Algorithms, providing detailed, actionable feedback on assignments for 30+ students and hosting weekly office hours

Cashier and Food Preparer: The Grand Beach Café - Wildwood, NJ — May 2023 - Aug 2025
- Served 100+ customers daily and trained 10+ new employees on procedures and POS software

PROJECTS

Carbon Calculus™ – Web Development, HTML/CSS, ReactJS, Vite, PlotlyJS, Git, GitHub Pages — Jun 2025 - Present
- Selected by a Sustainable Engineering PhD student to build an interactive site showcasing CO_2 research
- First undergraduate project funded by Villanova's Strategic Initiative for Climate, Justice, & Sustainability

American Sign Language Classifier – Machine Learning, Python, Scikit-Learn, PyTorch, Git — Aug 2025 - Dec 2025
- Trained a convolutional neural network to classify 36 numbers/letters in American Sign Language
- Documented end-to-end reproducibility in GitHub repository for model with ~95% accuracy

Kelly Criterion Simulation – Web Development, HTML/CSS, ReactJS, Git, GitHub Pages — May 2024 - Jul 2024
- Built and deployed an interactive Kelly Criterion simulation web app used by 15+ workshop participants, enhancing engagement and understanding of applied probability

RESEARCH PUBLICATIONS

#Haul: Analyzing YouTube Discourse on Over-Consumption & Sustainability — Jan 2024 - May 2024
DOI: 10.1145/3679318.3685373 – Published in NordiCHI 2024 Proceedings (Oct 2024)
- Implemented Python scripts and API integration to collect/analyze large-scale YouTube datasets with 1,000 videos, applying data modeling, text analysis, and visualization techniques

Non-Recursive Solution of the Tower of Hanoi — Jan 2024 - May 2024
DOI: 10.1080/0025570X.2025.2485001 – Published in Mathematics Magazine (Jul 2025)
- Collaborated with other academics to construct the first non-recursive equation describing the Tower of Hanoi, created an interactive website using HTML, CSS, and JavaScript to demonstrate our findings

AWARDS AND HONORS

ICPC Programming Competition, Division 1: 2nd at Site, 20th in Mid-Atlantic Region — Nov 2025
Phi Beta Kappa, Honor Society: Selected Based on Academic Excellence and Leadership — May 2025
ICPC Programming Competition, Division 2: 1st at Site, 8th in Mid-Atlantic Region — Nov 2024
National Merit Scholar: Top 1% of American High School Students — May 2023

LEADERSHIP, TEAMS, AND CLUBS

Co-President: Girls Who Code, Villanova Chapter — Apr 2025 - Present
- Cultivate a community of 50+ supportive women through educational, social, and technical meetings

Co-Founder: Aspirations in Computing Philadelphia Affiliate — Jan 2025 - Present
- Organize annual NCWIT banquet for 20+ Philadelphia winners of Aspirations in Computing Award

Vice President: Programming Team — Sep 2024 - Present
- Lead weekly competitive programming practice sessions for 130+ members, focusing on algorithmic problem-solving in Python, C++, and Java through *Kattis* (Individually ranked #4 in Pennsylvania)

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, R, SQL, SML, JavaScript, HTML/CSS
- **Frameworks/Tools:** Git, GitHub, React, Plotly, React Native, Electron, Scikit-Learn, PyTorch, NumPy, Pandas
- **Areas:** AI/ML, Data Visualization and Analysis, Web Development, API Integration, Front-End Development